

## FEISTRITZ OB BLEIBURG, Austria

WMO: 112320

Lat: 46.559N

Lon: 14.767E

Elev: 1742

StdP: 13.79

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.2       | 16.0 | 7.2                           | 8.5 | 17.0 | 10.5 | 10.0 | 19.7 | 15.1                  | 30.8 | 13.7 | 29.8 | 2.6                   | 240  | 0.485 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 87.4            | 70.1 | 84.0 | 68.5 | 80.8 | 67.3 | 71.5                | 84.9 | 69.9 | 81.7 | 68.2 | 78.8 | 6.0                  | 110  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.8                            | 105.3 | 79.5 | 65.4 | 100.1 | 77.0 | 64.0 | 95.2 | 74.5 | 36.4          | 85.5 | 34.9 | 81.8 | 33.4 | 78.9 | 77.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |       |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |       |
| 14.1              | 12.1 | 10.6 | DB                         | 8.4 | 92.7               | 5.1 | 3.4                                                | 4.7 | 95.2       | 1.7 | 97.2       | -1.1 | 99.1       | -4.8 | 101.5 |
|                   |      |      | WB                         | 7.8 | 74.4               | 5.0 | 1.8                                                | 4.1 | 75.7       | 1.2 | 76.7       | -1.6 | 77.7       | -5.3 | 79.0  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.0        | 28.2  | 33.2  | 42.1  | 50.3  | 57.0  | 64.3  | 68.3  | 66.9  | 58.2  | 49.2  | 39.6  | 29.5  | (6)  |
|                                                                        | DBStd       | 15.13       | 6.38  | 6.63  | 6.64  | 6.31  | 6.01  | 5.95  | 5.63  | 5.56  | 6.08  | 6.54  | 6.72  | 6.12  | (7)  |
|                                                                        | HDD50       | 2537        | 677   | 472   | 254   | 73    | 11    | 0     | 0     | 0     | 6     | 96    | 314   | 634   | (8)  |
|                                                                        | HDD65       | 6168        | 1142  | 891   | 711   | 442   | 255   | 83    | 33    | 44    | 216   | 491   | 761   | 1099  | (9)  |
|                                                                        | CDD50       | 2162        | 0     | 0     | 8     | 81    | 229   | 429   | 566   | 524   | 252   | 70    | 3     | 0     | (10) |
|                                                                        | CDD65       | 320         | 0     | 0     | 0     | 0     | 8     | 62    | 134   | 103   | 13    | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 2996        | 0     | 0     | 0     | 9     | 115   | 661   | 1242  | 895   | 74    | 0     | 0     | 0     | (12) |
|                                                                        | CDH80       | 863         | 0     | 0     | 0     | 0     | 12    | 166   | 405   | 276   | 4     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 4.1         | 3.6   | 4.0   | 4.7   | 4.9   | 4.8   | 4.7   | 4.3   | 4.0   | 3.9   | 3.8   | 3.3   | 3.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 42.4        | 1.4   | 1.7   | 2.2   | 2.9   | 3.9   | 5.0   | 5.2   | 5.3   | 4.5   | 4.1   | 3.7   | 2.4   | (15) |
|                                                                        | PrecMax     | 54.3        | 4.0   | 5.8   | 4.5   | 6.2   | 6.0   | 9.3   | 8.5   | 9.3   | 7.5   | 9.8   | 7.8   | 5.9   | (16) |
|                                                                        | PrecMin     | 34.6        | 0.2   | 0.0   | 0.5   | 0.6   | 1.8   | 2.3   | 1.5   | 2.0   | 2.0   | 0.0   | 0.4   | 0.2   | (17) |
|                                                                        | PrecStd     | 4.6         | 0.9   | 1.5   | 1.2   | 1.3   | 1.4   | 1.7   | 2.0   | 1.7   | 1.5   | 2.2   | 1.9   | 1.5   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 49.2  | 55.9  | 67.0  | 75.0  | 81.7  | 89.5  | 92.1  | 93.1  | 79.7  | 71.5  | 57.1  | 47.1  | (19) |
|                                                                        |             | MCWB        | 41.9  | 45.3  | 52.1  | 59.7  | 65.5  | 69.6  | 72.1  | 73.1  | 66.6  | 61.0  | 51.6  | 43.8  | (20) |
|                                                                        | 2%          | DB          | 42.4  | 49.5  | 62.0  | 70.0  | 77.1  | 84.4  | 87.8  | 86.4  | 76.1  | 66.8  | 53.8  | 43.1  | (21) |
|                                                                        |             | MCWB        | 38.8  | 42.3  | 49.8  | 55.5  | 62.5  | 68.1  | 70.3  | 70.0  | 65.2  | 57.8  | 51.2  | 40.6  | (22) |
|                                                                        | 5%          | DB          | 39.0  | 45.7  | 58.2  | 66.6  | 73.5  | 81.0  | 84.7  | 82.5  | 72.3  | 62.9  | 51.5  | 39.6  | (23) |
|                                                                        |             | MCWB        | 36.7  | 39.8  | 47.8  | 54.3  | 60.3  | 66.7  | 69.0  | 68.7  | 63.1  | 56.2  | 49.2  | 37.6  | (24) |
|                                                                        | 10%         | DB          | 36.5  | 42.2  | 54.0  | 62.9  | 69.8  | 77.9  | 81.1  | 78.9  | 68.7  | 59.2  | 48.8  | 37.3  | (25) |
|                                                                        |             | MCWB        | 34.7  | 37.8  | 45.6  | 52.5  | 58.7  | 64.9  | 67.7  | 67.2  | 61.2  | 54.2  | 47.1  | 35.9  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 42.8  | 46.2  | 52.4  | 60.8  | 67.8  | 72.3  | 73.8  | 73.7  | 67.5  | 61.2  | 53.7  | 44.6  | (27) |
|                                                                        |             | MCDB        | 48.1  | 53.6  | 64.5  | 73.6  | 78.1  | 85.8  | 88.5  | 89.5  | 77.4  | 70.1  | 55.3  | 45.6  | (28) |
|                                                                        | 2%          | WB          | 40.2  | 43.2  | 50.2  | 57.1  | 63.9  | 69.8  | 71.7  | 71.4  | 65.8  | 58.7  | 51.4  | 41.1  | (29) |
|                                                                        |             | MCDB        | 42.3  | 48.8  | 60.7  | 67.0  | 74.6  | 81.2  | 85.3  | 84.8  | 74.7  | 65.6  | 53.2  | 42.4  | (30) |
|                                                                        | 5%          | WB          | 36.9  | 40.6  | 48.4  | 54.8  | 61.4  | 67.5  | 70.0  | 69.7  | 63.9  | 56.7  | 49.1  | 38.0  | (31) |
|                                                                        |             | MCDB        | 38.1  | 45.2  | 57.0  | 64.6  | 71.6  | 78.9  | 82.5  | 80.8  | 71.1  | 61.9  | 51.2  | 39.1  | (32) |
|                                                                        | 10%         | WB          | 35.0  | 38.2  | 46.1  | 52.9  | 59.4  | 65.6  | 68.2  | 67.8  | 61.9  | 54.7  | 47.1  | 36.0  | (33) |
|                                                                        |             | MCDB        | 36.2  | 41.6  | 52.9  | 61.4  | 68.6  | 76.3  | 79.1  | 77.5  | 67.6  | 58.8  | 48.9  | 37.0  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.4   | 12.8  | 18.5  | 18.4  | 19.2  | 20.2  | 21.1  | 20.9  | 16.7  | 15.7  | 8.8   | 8.3   | (35) |
|                                                                        |             | MCDBR       | 12.4  | 19.9  | 26.0  | 26.1  | 26.2  | 26.6  | 26.9  | 26.9  | 21.8  | 20.8  | 13.1  | 11.7  | (36) |
|                                                                        | 5% WB       | MCWBR       | 8.9   | 13.1  | 15.1  | 13.9  | 13.3  | 13.0  | 12.0  | 13.0  | 11.7  | 12.8  | 9.6   | 9.0   | (37) |
|                                                                        |             | MCWBR       | 10.3  | 18.6  | 24.6  | 22.8  | 24.3  | 24.9  | 25.3  | 24.9  | 20.1  | 19.1  | 11.8  | 10.1  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.302 | 0.330 | 0.373 | 0.422 | 0.424 | 0.431 | 0.430 | 0.428 | 0.400 | 0.374 | 0.338 | 0.303 | (40) |
|                                                                        |             | taud        | 2.419 | 2.347 | 2.279 | 2.180 | 2.222 | 2.246 | 2.251 | 2.273 | 2.322 | 2.370 | 2.425 | 2.472 | (41) |
|                                                                        | Ebn at Noon |             | 252   | 265   | 268   | 264   | 268   | 266   | 264   | 260   | 256   | 245   | 234   | 239   | (42) |
|                                                                        |             | Edh at Noon | 22    | 29    | 36    | 43    | 43    | 43    | 42    | 39    | 34    | 28    | 22    | 19    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 404   | 667   | 1033  | 1343  | 1630  | 1822  | 1824  | 1605  | 1144  | 755   | 415   | 332   | (44) |
|                                                                        | RadStd      |             | 57    | 100   | 133   | 152   | 158   | 123   | 126   | 173   | 150   | 85    | 66    | 50    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (89 neighbors) | N/A   | N/A     | N/A    | +1.53   | +0.83 | +0.63 | -153        | -241  | +166  | +103  |

Nomenclature: See separate page